Einstein realized that gravity and accelerated motion are two sides of the same coin.

Just as by changing your planned motion (to avoid accelerating) you can avoid feeling squeezed back in your car seat or feeling pushed sideways on the train, Einstein understood that by suitably changing your motion you can also avoid feeling the usual sensations associated with gravity's pull. The idea is wonderfully simple. To understand it, imagine that Barney is desperately trying to win the Springfield Challenge, a monthlong competition among all belt-size-challenged males to see who can shed the greatest number of inches. But after two weeks on a liquid diet (Duff Beer), when he still has an obstructed view of the bathroom scale, he loses all hope. And so, in a fit of frustration, with the scale stuck to his feet, he leaps from the bathroom window. On his way down, just before plummeting into his neighbor's pool, Barney looks at the scale's reading and what does he see? Well, Einstein was the first person to realize, and realize fully, that Barney will see the scale's reading drop to zero. The scale falls at exactly the same rate as Barney does, so his feet don't press against it at all. *In free fall, Barney experiences the same weightlessnessthat astronauts experience in outer space.*

Indeed, if we imagine that Barney jumps out his window into a large shaft from which all air has been evacuated, then on his way down not only would air resistance be eliminated, but because every atom of his body would be falling at exactly the same rate, all the usual external bodily stresses and strains—his feet pushing up against his ankles, his legs pushing into his hips, his arms pulling down on his shoulders—would be eliminated as well.[14] By closing his eyes during the descent, Barney would feel exactly what he would if he were floating in the darkness of deep space. (And, again, in case you're happier with nonhuman examples: if you drop two rocks tied by a rope into the evacuated shaft, the rope will remain slack, just as it would if the rocks were floating in outer space.) Thus, by changing his state of motion—by fully "giving in to gravity"— Barney is able to simulate a gravity-free environment. (As a matter of fact, NASA trains astronauts for the gravity-free environment of outer space by having them ride in a modified 707 airplane, nicknamed the *Vomit Comet,* that periodically goes into a state of free fall toward earth.)